

Health and economic burden of benign gastrointestinal conditions

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Outline

- Prevalence of common GI conditions
- Definitions
- Common conditions
 - Irritable Bowel Syndrome (IBS)
 - Inflammatory Bowel Disease (IBD)
- Diagnosis
- Impact

Prevalence of GI conditions in the UK

Condition	Prevalence	Incident cases/year
Crohn's Disease	0.15%	3000-6000
Ulcerative colitis	0.2%	6000-12000
Coeliac Disease	1%	7000
IBS	10.5%	78,600
Colorectal Cancer	5%	33,000

Definitions

- Intestine = stomach + small bowel + large bowel
- Small bowel = duodenum + jejunum + ileum
- Large bowel = colon
- Acute < 6 weeks Chronic > 6 weeks
- Endoscopy = Gastroscopy + Colonoscopy + Flexible Sigmoidoscopy
- IBD (inflammatory bowel disease) = ulcerative colitis + Crohn's disease

Irritable Bowel Syndrome - IBS

- On-going abdominal symptoms in the absence of any demonstrable pathology
- 40% of patients referred to secondary care GI clinics

Background

Irritable bowel syndrome (IBS) has a prevalence of 10-20% in the general population

It is a chronic, relapsing and often life-long disorder

The people most commonly affected are those aged 20–30 years

It is twice as common in women as in men

IBS symptoms

Consider IBS if any of these symptoms have been present for at least 6 months

- Abdominal pain or discomfort
- Bloating
- Change in bowel habit

IBS Diagnostic criteria

ROME III¹

Recurrent abdominal pain/discomfort for >3 days/month in the past 3 months, associated with two or more of:

- Improvement with defaecation
- Onset associated with change in stool frequency
- Onset associated with change in stool form

NICE²

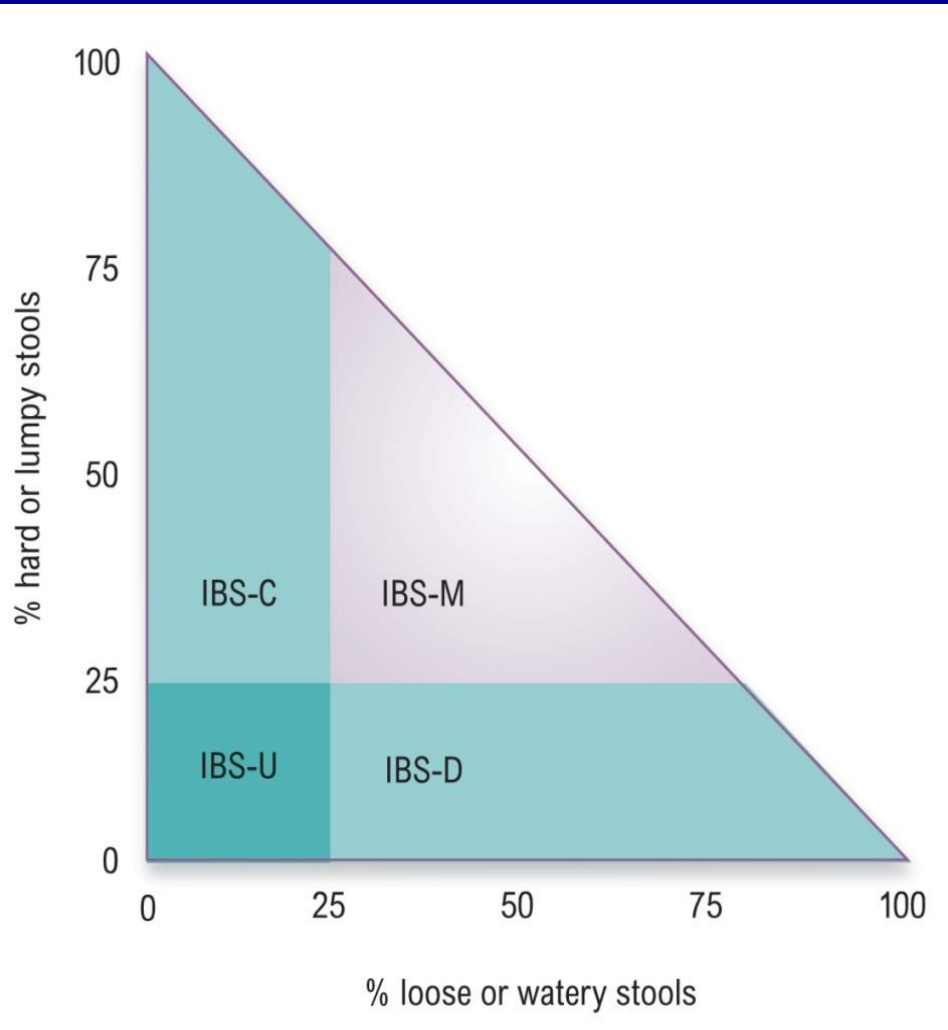
Abdominal pain/discomfort relieved by defaecation or associated with altered stool frequency/form, plus two or more of:

- Altered stool passage
- Abdominal bloating/distension
- Symptoms made worse by eating
- Passage of mucus

Investigations in IBS

- Bloods
- Colonoscopy
- Imaging

Three major IBS subtypes



IBS with constipation: >25% hard stools, <25% loose stools

IBS-mixed: both hard and loose stools

IBS with diarrhoea: >25% loose stools, <25% hard stools

IBS Disease Burden

- **Social**
 - Restrict social activities
 - Reduce quality of life
- **Economic**
 - Work days lost in USA – 14.8 compared to 8.7 in normal population¹
- **Healthcare System**
 - High prevalence with recurrent symptoms – increased use of resources
 - 40% patients seen in GI OPD²

¹ Drossman DA *et al.* *Dig Dis Sci* 1993; 38:1569-80

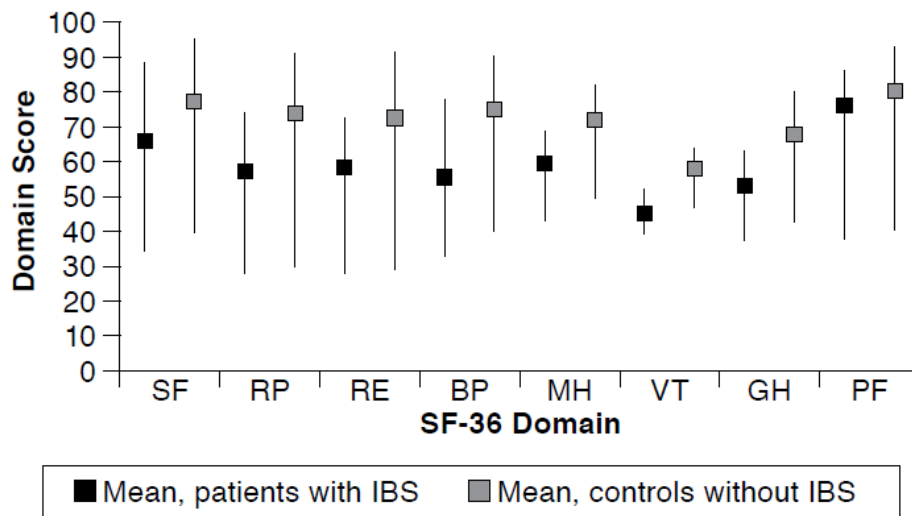
² Mitchell CM *et al.* *Gastroenterology* 1987; 92:1282-4

A Systematic Review of the Economic and Humanistic Burden of Illness in Irritable Bowel Syndrome and Chronic Constipation

Dave Nellesen, PhD, MBA; Kimberly Yee, MPH; Anita Chawla, PhD;
Barbara Edelman Lewis, PhD, MHA; and Robyn T. Carson, MPH

FIGURE 2

Range of SF-36 Domain Scores for IBS Patients Compared with Non-IBS Controls (14 Studies)



BP=bodily pain; GH=general health; IBS=irritable bowel syndrome; MH=mental health; PF=physical functioning; RE=role emotional; RP=role physical; SF=social functioning; VT=vitality.

Economic Burden of Irritable Bowel Syndrome with Constipation: A Retrospective Analysis of Health Care Costs in a Commercially Insured Population

Jalpa A. Doshi, PhD; Qian Cai, MS, MSPH; Jessica L. Buono, MPH; William M. Spalding, MS; Michael Sarocco, RPh, MS; Hiangkiat Tan, MS, BPharm; Judith J. Stephenson, SM; and Robyn T. Carson, MPH

- IBS-C v Healthy controls
- Annual cost : IBS \$8,621 Controls \$4,765 ($p < 0.01$)
- Costs mainly through increased clinic visits

Burden of irritable bowel syndrome in an increasingly cost-aware National Health Service

Anet Soubieres,¹ Patrick Wilson,¹ Andrew Poullis,¹ Julia Wilkins,² Mark Rance³

ICD-10 code	N	Per cent	Cost	Per cent of overall cost
IBS specific				
IBS with diarrhoea (K58.0)	918	0.8	£824 897	0.9
IBS without diarrhoea (K58.9)	2599	2.2	£1 959 408	2.0
Total	3517	3.0	£2 784 305	2.9
IBS-related symptoms				
Constipation (K59.0)	23 529	20.2	£24 526 655	25.6
Change in bowel habit (R19.4)	15 986	13.7	£8 756 006	9.2
Abdominal pain (R10.4)	73 275	63.0	£59 625 103	62.3
Total	112 790	97.0	£92 907 764	97.1

IBS Summary

- IBS is common
- IBS is expensive
- IBS affects QOL

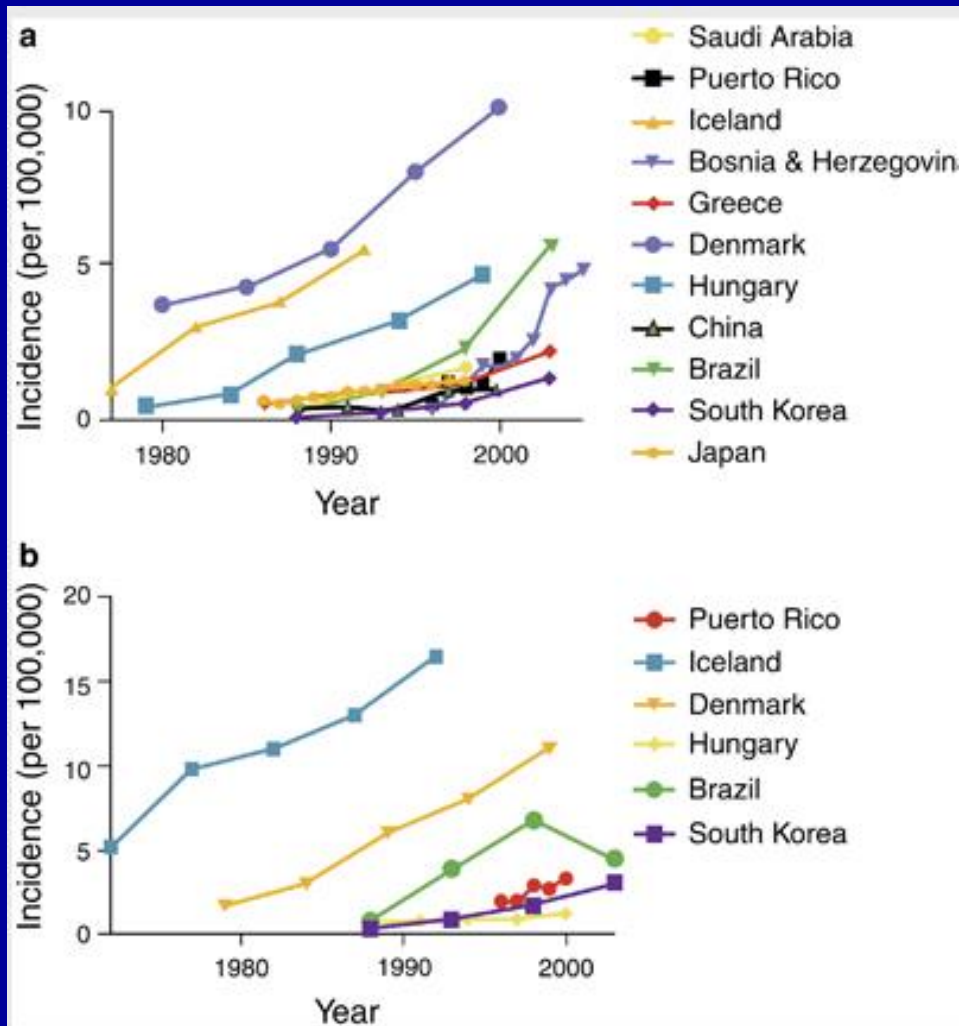
Inflammatory Bowel Disease (IBD)

- Long term conditions
- Remission and exacerbations
- Crohn's disease and Ulcerative colitis

Definitions

- Acute / Chronic
- Severity (mild/moderate/severe)
- Relapse / remission
- Quiescent

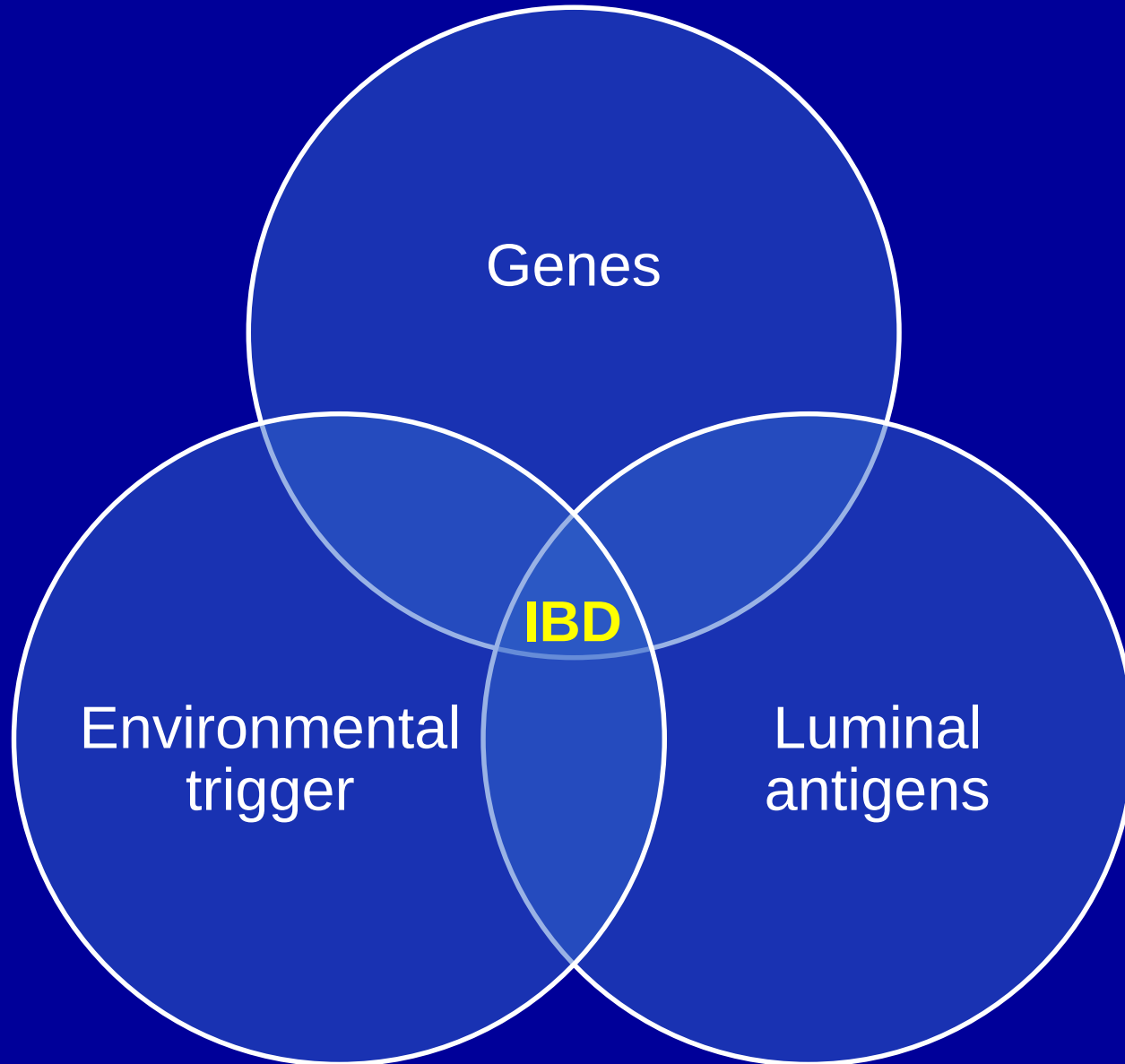
Incidence of IBD

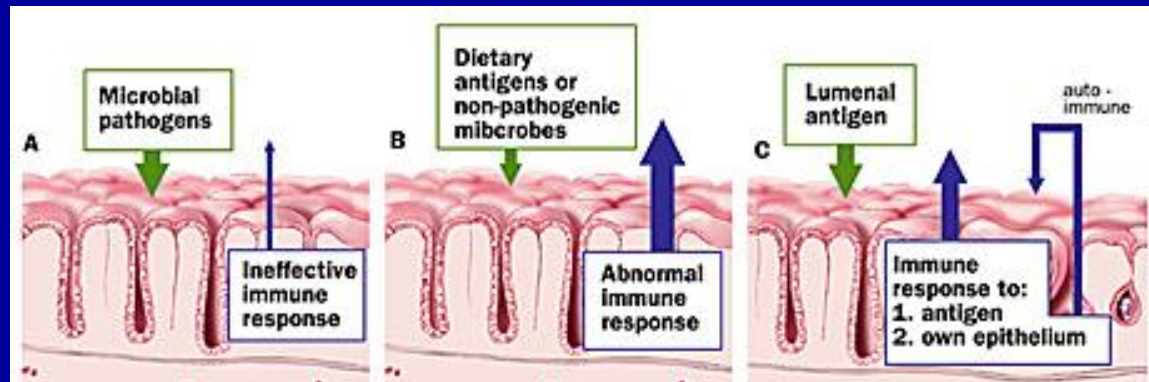
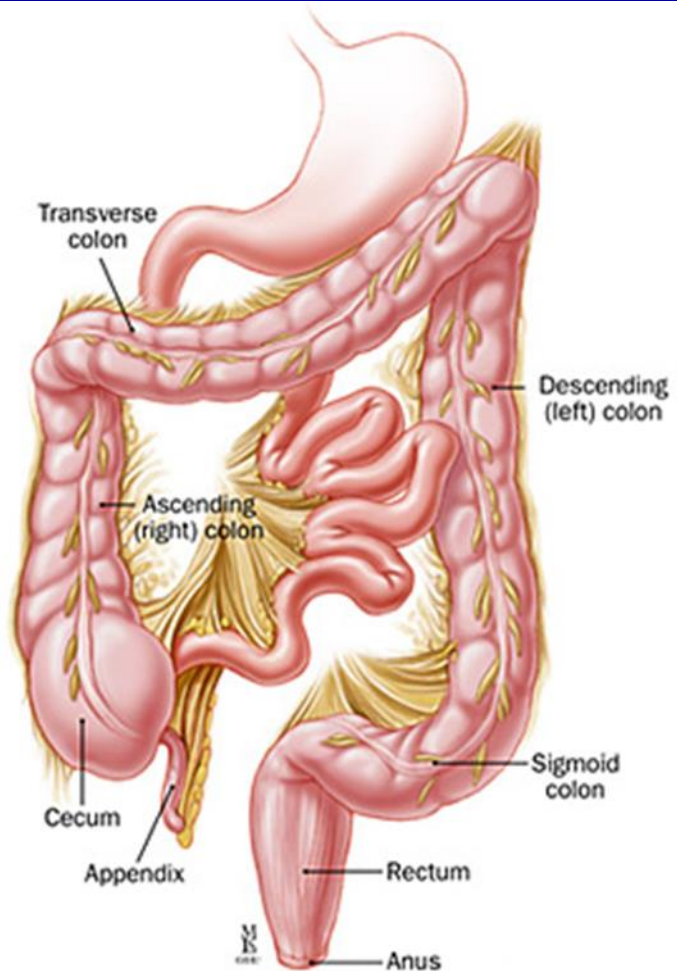


CD

UC

Cause of IBD





Genetics / Inheritance

- Increased risk of CD and UC in offspring if parent has IBD¹
 - 5% if 1 parent has CD
 - 1.6% if 1 parent has UC
- If both parents have IBD risk could be as high as 35% ²
- Breast feeding may reduce risk ³

1 Orholm et al Am J Gastroenterolo 1999;94:3236-8

2 Bennett et al, Gastroenterology 1991;100:1638-43

3 Yang et al Gut 1993;34:517-24

Symptoms

- Diarrhoea (often bloody)
- Abdominal pain
- Fatigue + malaise



Eyes
Episcleritis
Uveitis



Kidneys
Stones
(nephrolithiasis)
Hydronephrosis
Fistulae
Urinary tract
infection

Skin
Erythema nodosum
Pyoderma
gangrenosum



M
K
GNE



Mouth
Stomatitis
Aphthous ulcers



Liver
Steatosis

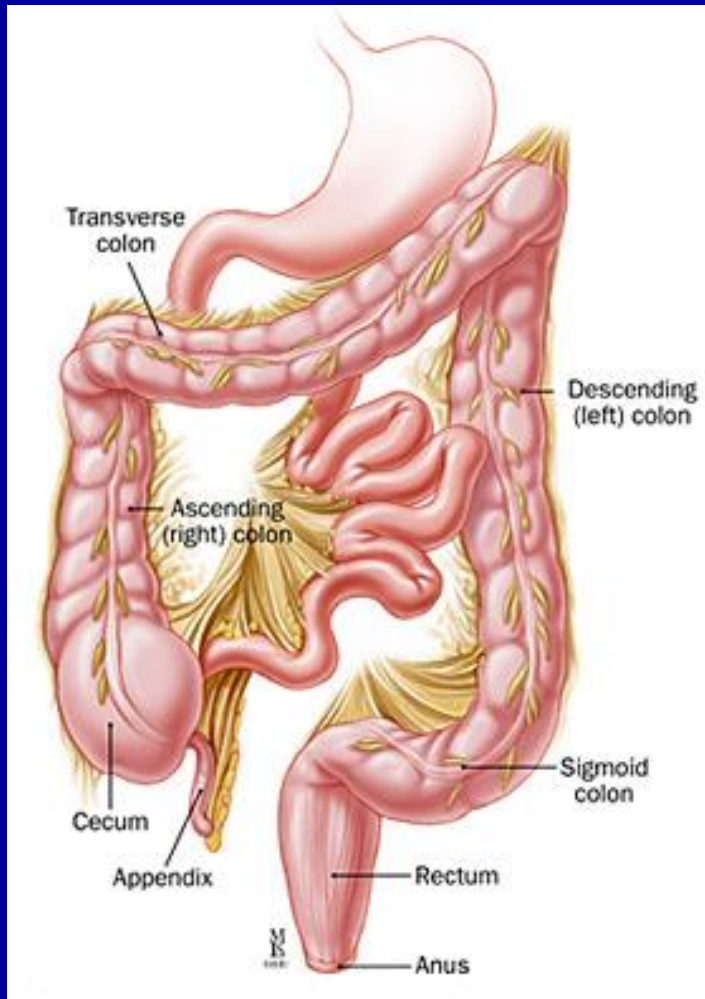
Biliary tract
Gallstones
Sclerosing cholangitis

Joints
Spondylitis
Sacroiliitis
Peripheral arthritis

Circulation
Phlebitis

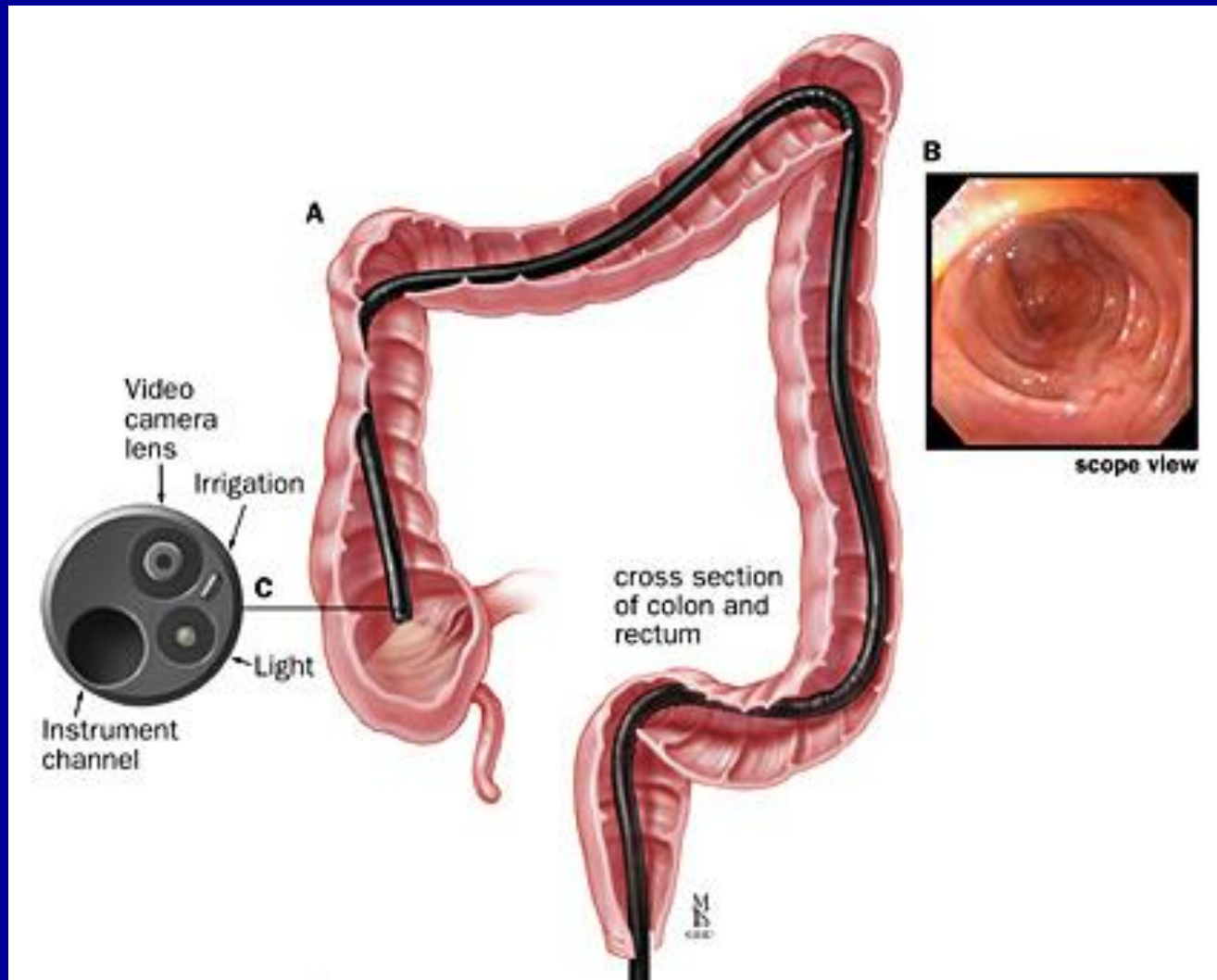
Extra-intestinal manifestations of IBD

Diagnosis

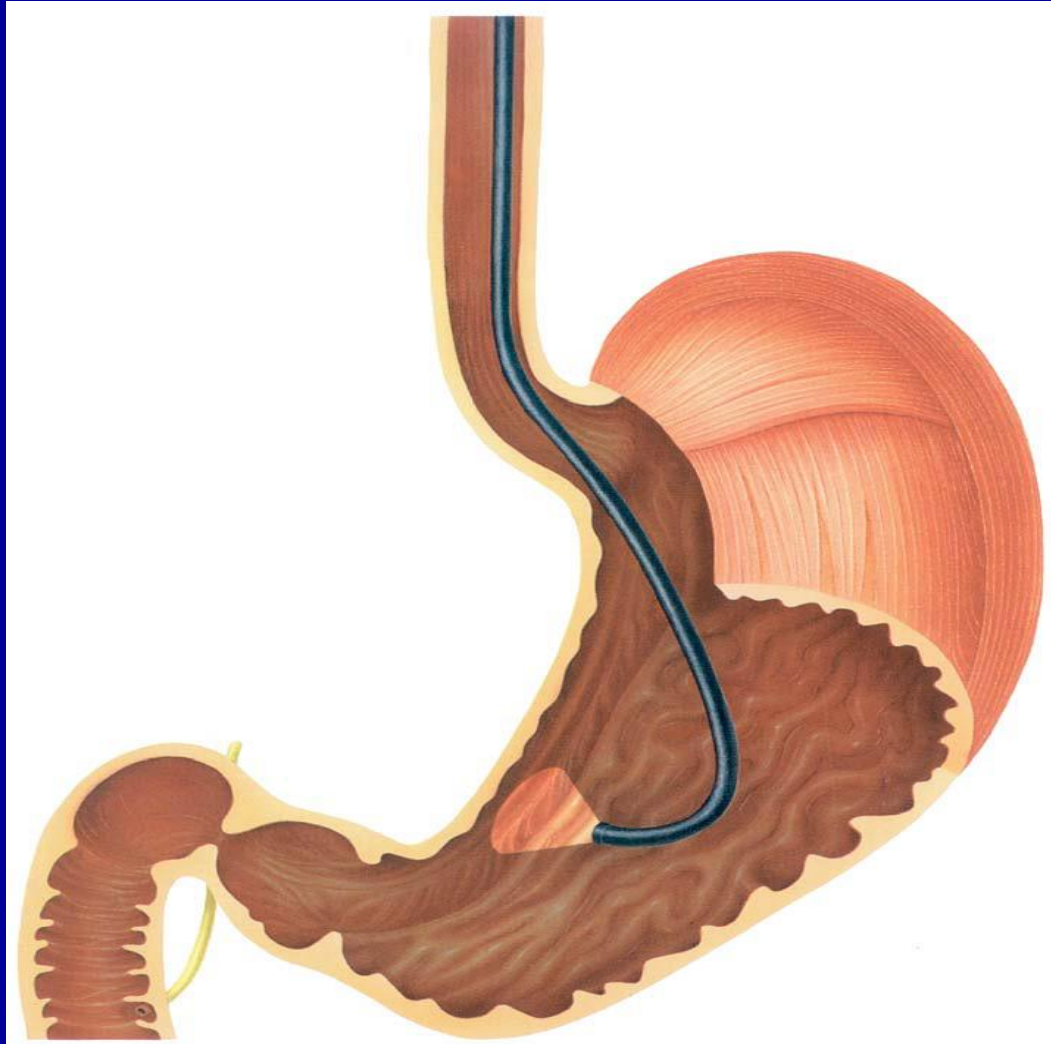


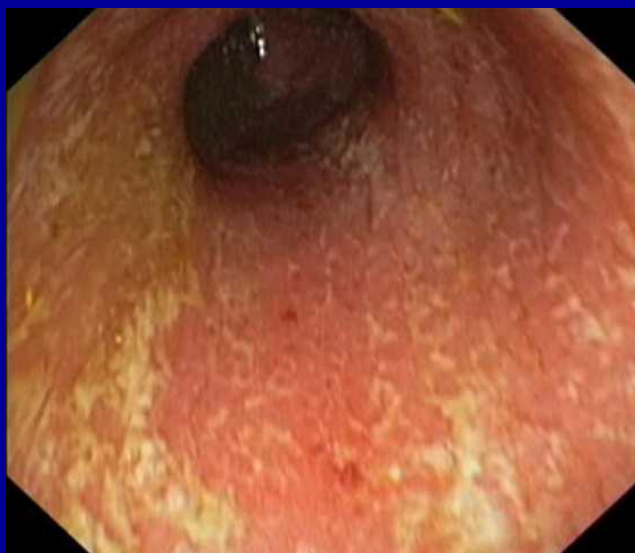
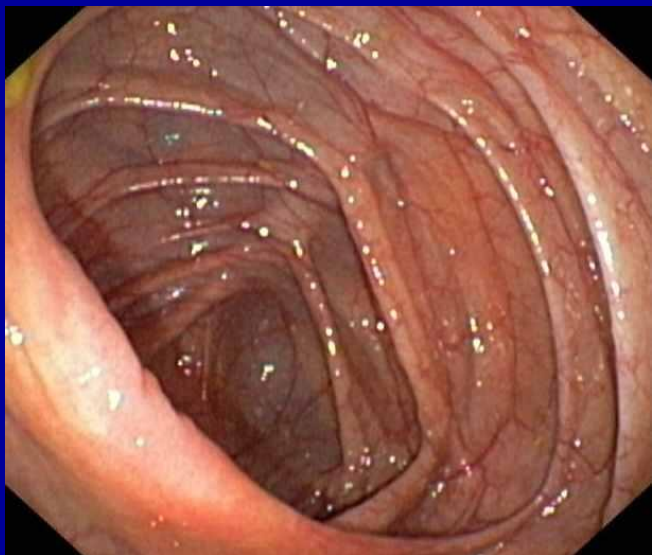
- Blood tests
- Stool tests
- Radiology
- Endoscopy

Colonoscopy



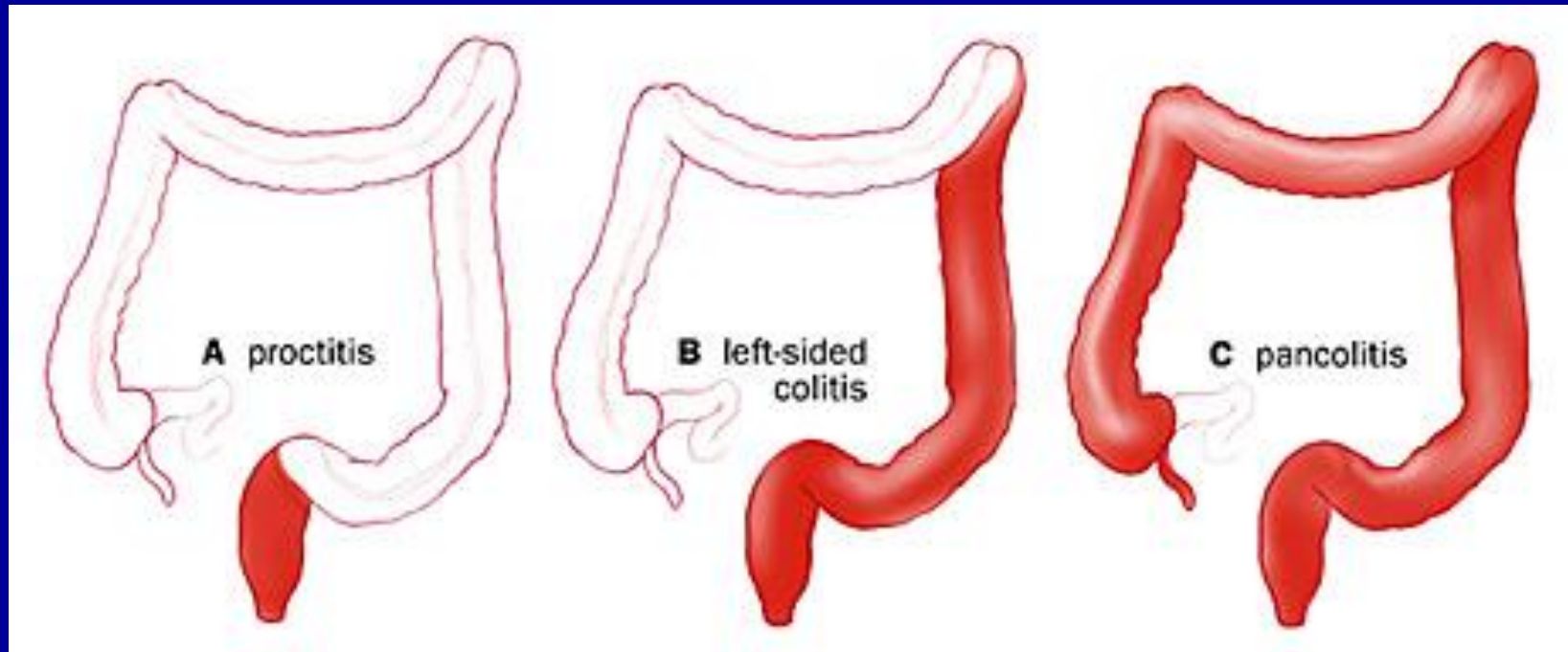
Gastrosocopy





Ulcerative colitis

Ulcerative colitis (UC)



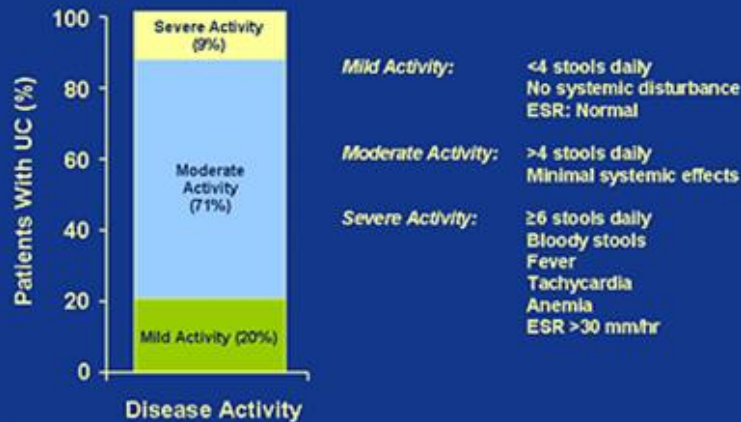
Montreal classification of ulcerative colitis

Extent		Anatomy
E1	Ulcerative proctitis	Rectum only
E2	Left sided (distal)	Distal to splenic flexure
E3	Extensive (pan-colitis)	Proximal to splenic flexure

Mayo Clinic score		score
Stool Frequency	Normal number for patient	0
	1-2 more than normal	1
	3-4 more	2
	>=5 more	3
Rectal bleeding	None	0
	Steaks < 50% of time	1
	Most of the time	2
	Blood alone	3
Endoscopic activity	None	0
	Mild	1
	Moderate	2
	Severe	3
Physicians Global assessment	Normal	0
	Mild	1
	Moderate	2
	Severe	3

Initial Natural History of UC

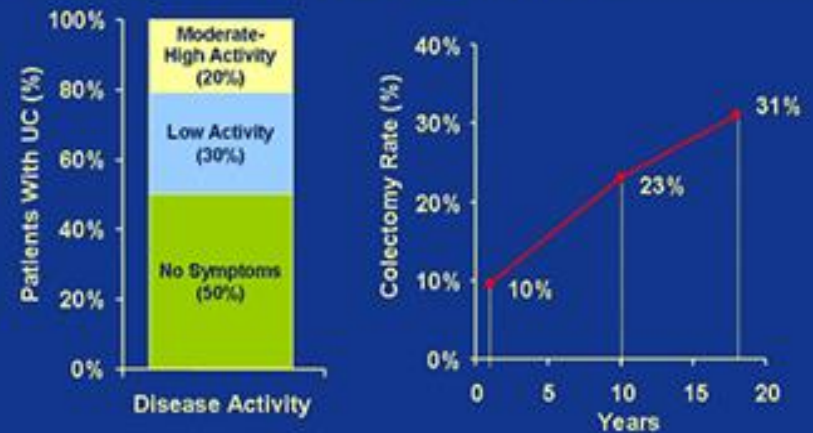
Disease Severity at Presentation



ESR = erythrocyte sedimentation rate

Source: Hendriksen C, et al. Gut. 1985;26:158-163.

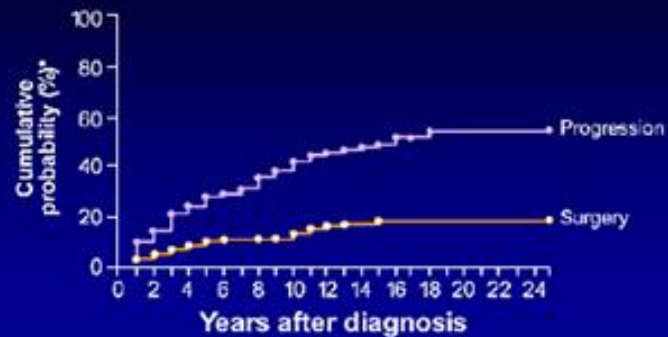
Disease course 1 year after diagnosis



Source: Hendriksen C, et al. Gut. 1985;26:158-163.

Longer term Natural History

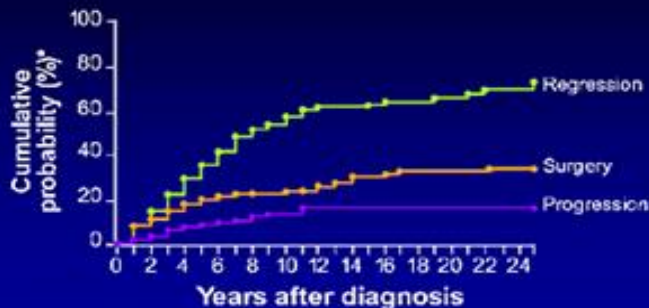
Natural Course of UC: Proctosigmoiditis



*Based on a multivariate analysis.

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Scand J Gastroenterol. 1996;31:260-266.

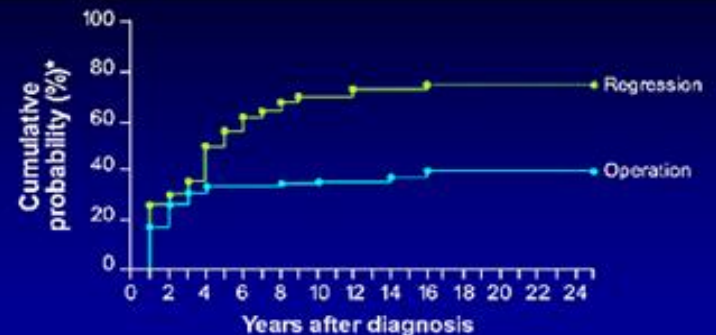
Natural Course of UC: Substantial Colitis (Left-sided and Extensive)



*Based on a multivariate analysis.

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Scand J Gastroenterol. 1996;31:260-266.

Natural Course of UC: Pancolitis



*Based on a multivariate analysis.

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Langholz E et al. Scand J Gastroenterol. 1996;31:260-266.

Acute severe ulcerative colitis

- Mortality untreated 24%
 steroids 7%
 timely surgery <1%

Truelove and Witts *BMJ* 1955

Rice-Oxley and Truelove *Lancet* 1950

But.....

- 2001 Audit suggested D.G.H. mortality might be as high as 24%
 - Delays in surgery
 - Delays in diagnosis
 - Undue persistence with unsuccessful medical treatment

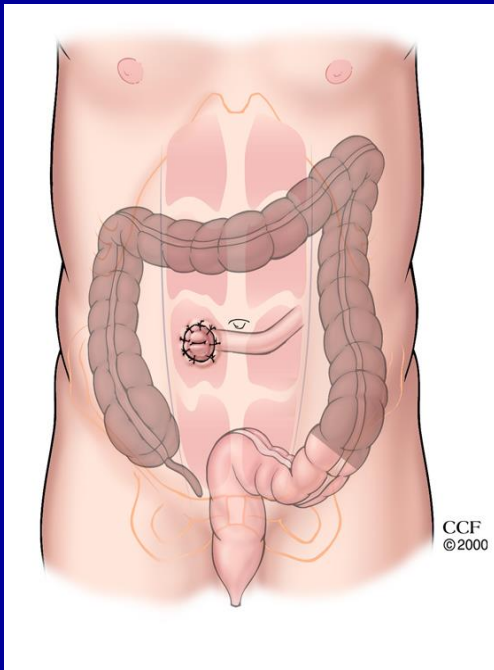
Treatments

- Drugs
 - 5-aminosalicylates
 - Steroids
 - Immunomodulators
 - Biologics
 - anti TNF
 - $\alpha 4\beta 7$ integrans
- Surgery

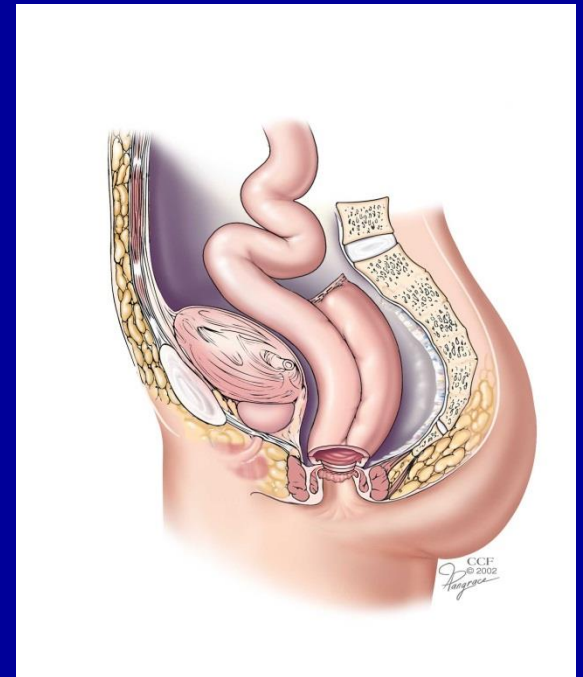
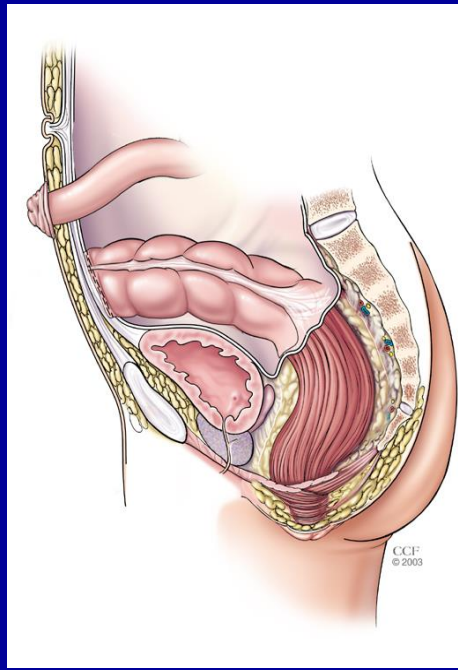
Ulcerative Colitis



Staged surgical procedures



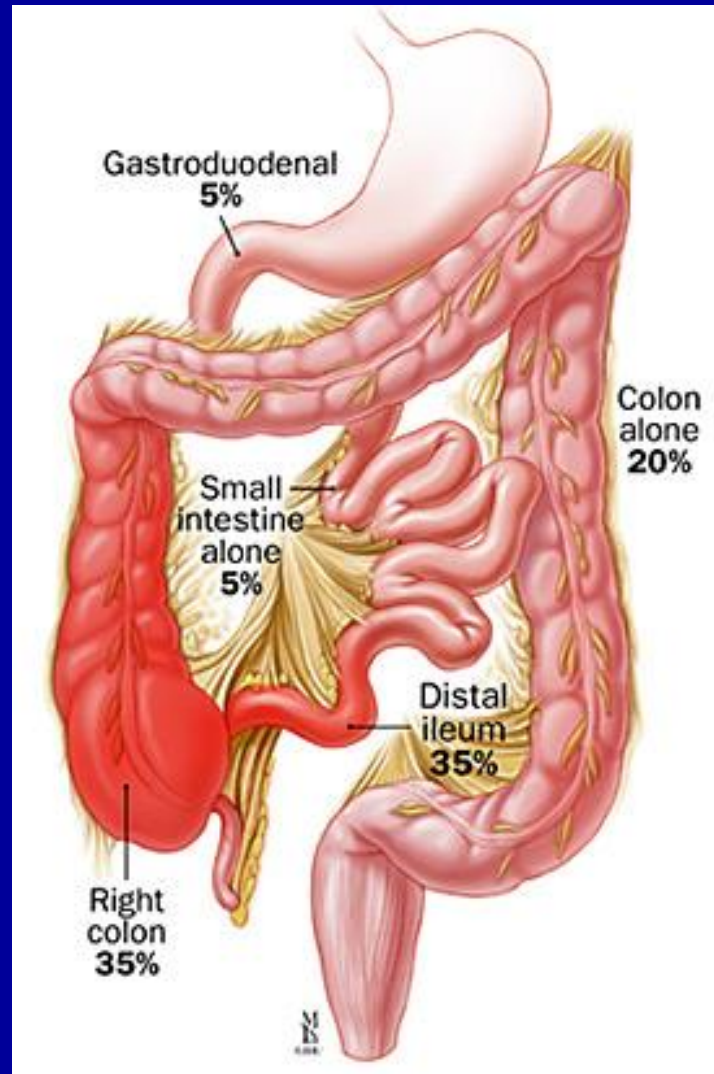
Colectomy + ileostomy



Ileoanal Pouch

Crohn's Disease

Crohn's Disease



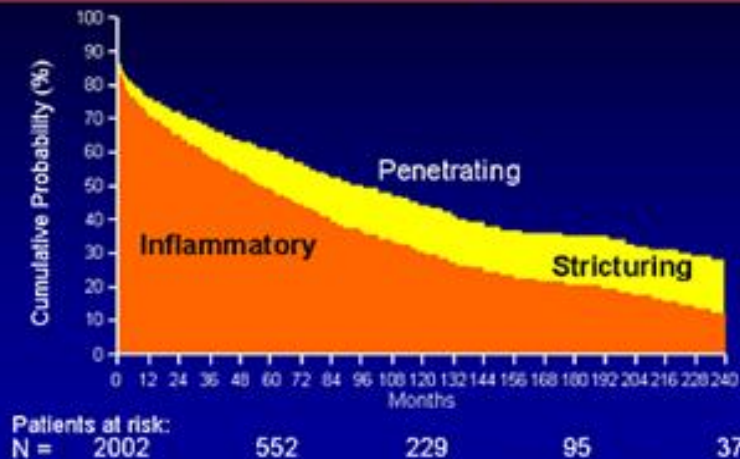
Montreal Classification of Crohn's

Age	A1 <16 years
	A2 17-40
	A3 >40
Location	L1 ileal
	L2 colonic
	L3 ileocolonic
	L4 Isolated upper GI disease
Behaviour	B1 non-stricturing, non-penetrating
	B2 stricturing
	B3 penetrating
	P peri-anal disease (added to B1-3 when present)

Harvey Bradshaw Index		score
General well being	Very well	0
	Slightly below par	1
	Poor	2
	Very poor	3
	Terrible	4
Abdominal pain	None	0
	Mild	1
	Moderate	2
	Severe	3
Number of liquid stools		
Abdominal mass	None	0
	Dubious	1
	Definite	2
	Definite and tender	3
Complications	Add 1 per complication	

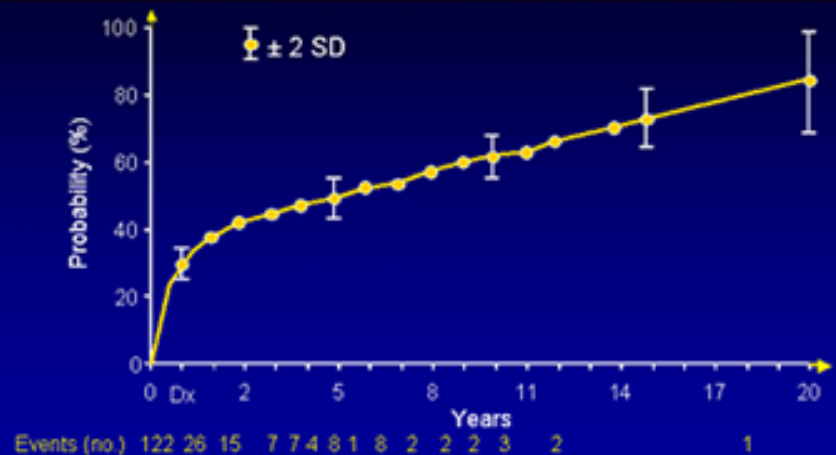
Natural History of Crohn's

Long-Term Evolution of Disease Behavior in CD



Cosnes J et al. *Inflamm Bowel Dis*. 2002;8:244.

Cumulative Probability of Surgical Intervention in CD



Munkholm P et al. *Gastroenterology*. 1993;105:1716.

Treatments

- Drugs
 - Steroids
 - Immunomodulators
 - Biologics
 - anti TNF
 - $\alpha 4\beta 7$ integrans
- Surgery

Mortality in Crohn's

- Survival is modestly reduced (6-7% over 20 years)
- Females diagnosed before the age of 50
- Disease-related complications, gastrointestinal cancer



IBD summary

- IBD is becoming more common
- Progressive changes to treatments has improved outlook
- IBD is a dynamic and variable condition

Questions ?